

Full Stack Development with MERN

Functional Specification Document

STOCK TRADING APP

1. Introduction

Project Title: STOCK TRADING APP

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2. Project Overview

Purpose: SB Stocks is a full-stack stock trading simulation platform for practicing stock trades with virtual funds.

Features include authentication, dashboard, stock search, quote/history, buy/sell simulation, transactions, watchlist, admin stock management, and responsive UI.

3. Architecture

Frontend: React, React Router, Tailwind CSS, Recharts.

Backend: Node.js, Express.js, controllers, routes, middleware, and services.

Database: MongoDB Atlas with Mongoose models for User, Holding, Transaction, Watchlist, and Stock.

4. Setup Instructions

Prerequisites: Node.js 20+, npm, MongoDB Atlas, Finnhub API key, Git.

Installation: clone repo, copy server/.env.example, set variables, run npm run install:all, npm run seed, and npm run dev.

5. Folder Structure

Client: client/src/components, pages, hooks, services.

Server: server/src/config, controllers, middleware, models, routes, services.

6. Running the Application

Local frontend: http://localhost:5173

Local backend: <http://localhost:5000>

Live demo: <https://sb-stocks-docker.onrender.com>

7. API Documentation

Method	Endpoint	Description
POST	/api/auth/register	Register a new user account.
POST	/api/auth/login	Authenticate user and return JWT token.
GET	/api/auth/me	Return logged-in user profile.
GET	/api/stocks/search?q=	Search stocks by ticker or company name.
GET	/api/stocks/quote/:ticker	Return current quote for a ticker.
GET	/api/stocks/history/:ticker?range=	Return historical price data.
POST	/api/trade/buy	Buy shares using virtual cash.
POST	/api/trade/sell	Sell shares from existing holdings.
GET	/api/portfolio	Return cash, holdings, values, and gain/loss.
GET	/api/transactions	Return trade transaction history.
GET/POST	/api/watchlist	View or add watchlist tickers.
DELETE	/api/watchlist/:ticker	Remove ticker from watchlist.
GET/POST	/api/admin/stocks	Admin stock listing and creation.
PUT/DELETE	/api/admin/stocks/:id	Admin stock update and delete.

8. Authentication

JWT tokens are used for sessions. Passwords are hashed with bcrypt. Protected routes use `requireAuth`, and admin APIs use `requireAdmin`.

9. User Interface

Pages include Login, Register, Dashboard, Stock Detail, Transactions, Profile, Watchlist, and Admin Stocks.

10. Testing

Tested authentication, protected routes, trading validation, portfolio updates, transaction history, and Render health checks.

11. Screenshots or Demo

GitHub Repository: https://github.com/duday8329-ai/stock_trading_app

Live Demo: <https://sb-stocks-docker.onrender.com>

Screenshots can include login, dashboard, stock detail, transaction history, admin panel, Render, and MongoDB Atlas.

12. Known Issues

Render free service can spin down with inactivity. Free-tier market data can be delayed or rate-limited.

13. Future Enhancements

Leaderboards, multiple virtual portfolios, price alerts, dark mode, and advanced portfolio analytics.